

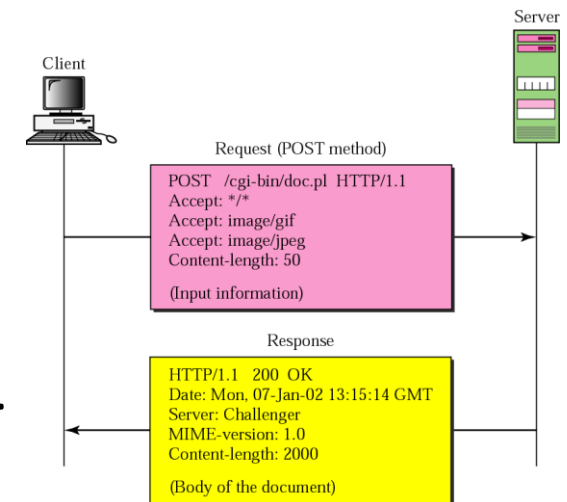
TUTORIAL – 2

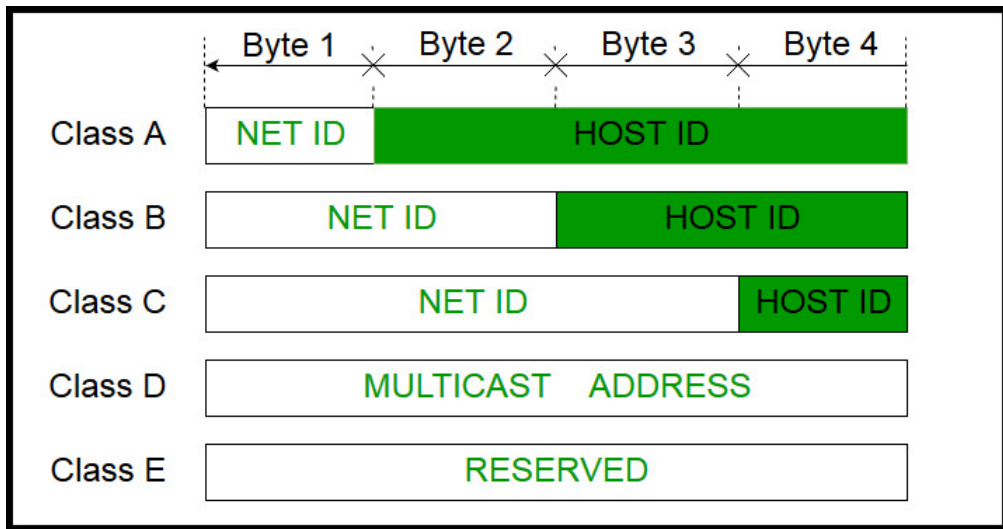
Q1. Assume that the IP address of a network device is 141.14.31.29,

- (1) Which **class** does the IP Address belong to?
- (2) What are the **netid** and the **hostid** of the IP address?
- (3) What is the **default mask** for the network?
- (4) In this case, how do you determine the **netid** with the **default mask**?
- (5) If the device sends a packet to all hosts in the network, what is the **source and destination IP addresses** in the packet?

Q2. A HTTP transaction is shown as follows:

- (1) What is the request **method**? What is **URL**?
- (2) What is the **status code** and **phrase**?
- (3) List the **request headers** and the **response headers** and their **values**.
- (4) List the general headers and entity headers and their values if any.





Class	IP address range (1 st Octet)	Network Mask	Prefix	Number of Networks	Number of Hosts
A	1. - 127.	255.0.0.0	/8	125	16,777,214
B	128. - 191.	255.255.0.0	/16	16,382	65,534
C	192. - 223.	255.255.255.0	/24	2,097,150	254
D	224. - 239.	Multicast addresses			
E	240. - 254.	Restricted/Experimental			

Example: IP=134.45.78.2 is a class B (128-191) with Netid=134.45, so its network address is 134.45.0.0

**** IP AND DM = NID ** Source: 134.45.78.2 Dest: 134.45.255.255**

An IPv4 address (dotted-decimal notation)

172 . 16 . 254 . 1

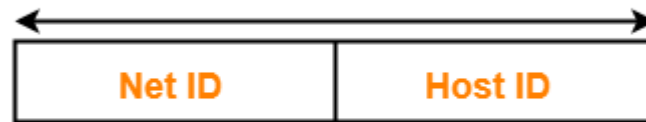
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10101100 . 00010000 . 11111110 . 00000001

One byte = Eight bits

Thirty-two bits (4 x 8), or 4 bytes

32 bits



Format of an IP Address

IP: 169 . 174 . 141 . 10

Subnet: 255 . 255 . 255 . 240

IP Address: 10101001 10101110 10001101 00001010

Subnet mask: 11111111 11111111 11111111 11110000

Network ID: 10101001 10101110 10001101 00000000

Network IP: 169 . 174 . 141 . 0

$$2^4 = 16$$

Number of hosts: 16

Number of usable hosts: 14

Usable hosts: 169.174.141.1 - 169.174.141.14

S.No	IP Address	Class	Net id	Host id
1.	227.34.78.7	Class D	no net id	no host id
2.	4.23.145.90	Class A	4	23.145.90
3.	198.76.9.23	Class C	198.76.9	23
4.	129.6.8.4	Class B	129.6	8.4
5.	246.7.3.8	Class E	no net id	no host id

A = 146.13.31.29 [X.Y.P.Q]				
Class	netId/hostId	Default Mask	Calculate NetID	Source/Destination
	X.Y.0.0/0.0.P.Q	255.255.0.0 = B	A AND B	X.Y.P.Q / X.Y.255.255

Method	URL	Status Phrase	Request	Response	Entity
post	/cgi-bin/doc.pl	200 ok	Accept content	Challenge r	Cont. Length, Mime

